Technical Appendix: Regulatory Separation and Pharmaceutical Innovation: Evidence from China's MAH Reform

Contents

1 Appendix Roadmap and Scope

This appendix is the formal derivation layer for the MAH model. It defines every primitive before use, derives product pricing and route values from explicit technologies, proves deterministic organizational sorting, solves the common project-advancement problem, closes the single qualified manufacturing-service market, and records all six required propositions plus the novelty corollary. It then separates project advancement, organizational assignment, observed holder–producer separation, and realized products. The revised baseline also distinguishes predetermined commercialization financing capacity from internal manufacturing capability and treats route financing requirements as feasibility thresholds rather than duplicated costs.

The baseline has one binary policy channel, $M \to \tau_E(M)$, one common control x_i, deterministic choice among I, E, T, A, route-independent downstream success, and one scalar qualified-capacity equilibrium. It contains no entry, product-market, labor, capital, portfolio, dynamic-firm, or welfare closure. The final sections document extensions behind an explicit strict separation from the baseline. Research–development allocation, patent production, borrowing, and a finance market remain inactive.

2 Primitives, Objects, and Timing

This section fixes the objects and information structure of the baseline before any payoff is constructed. The model is a partial equilibrium of original-drug project advancement, commercialization organization, and qualified manufacturing services. The MAH reform changes only the institutional feasibility or friction of retained entrusted manufacturing.

2.1 Units and populations

Let C denote currency units, X project-advancement input units, P the expected measure of viable planning-stage projects in one decision cohort, K manufacturing capability/requirement units, and B qualified manufacturing-service capacity units. Commercialization financing capacity and requirements are one-time monetary commitments measured in C, but they are stocks of fundable liquidity rather than additional payoff costs.

There is a continuum $\mathcal{I}$ of original-drug developers. Developer characteristics follow the exogenous joint distribution $H(a, k, \ell)$, with no independence restriction. Developer i has research capability $a_i > 0$, measured in P/X, and internal manufacturing capability $k_i > 0$, measured in K. It also has predetermined commercialization-stage financing capacity $\ell_i > 0$, measured in C. Financing capacity is the largest up-front organizational commitment the developer can fund; it is neither research productivity nor manufacturing know-how. Its complete permanent baseline characteristic vector is

$$\theta_i = (a_i, k_i, \ell_i). \tag{1}$$

These are continuous characteristics, not permanent discrete firm types.

There is also a set $\mathcal{J}$ of qualified manufacturing-service suppliers. Supplier efficiency $z_j > 0$ follows the exogenous distribution $H_C(z)$. The supplier's capacity control and cost function are introduced when the qualified manufacturing-service market is defined below.

2.2 Project characteristics and empirical classification

A viable project that reaches the route-planning stage draws commercial or scientific value shifter $q > 0$ and manufacturing requirement $m > 0$. The latter is measured on the same K scale as k_i. Define

$$\omega = (q, m) \sim F(q, m), \tag{2}$$

where F is an exogenous joint distribution. Thus q governs the project's value side, while the pair (m, k_i) governs the manufacturing mismatch. The institutional regime does not directly alter q, m, or F.

For empirical decompositions only, let $g \in \{O, \mathrm{Inc}\}$ be an exogenous novelty classifier. The label Inc cannot be confused with internal route I. Let $\rho_g \geq 0$ be the class share and $F_g(q, m)$ the conditional project distribution. They satisfy

$$\sum_{g \in \{O, \mathrm{Inc}\}} \rho_g = 1, \qquad F(q, m) = \sum_{g \in \{O, \mathrm{Inc}\}} \rho_g F_g(q, m). \tag{3}$$

The classifier is not a new baseline state, and it does not create a separate control x_{ig}.

2.3 Institutional regime and route labels

Let the exogenous institutional regime be $M \in \{0, 1\}$. The value $M = 0$ denotes the pre-MAH regime. The value $M = 1$ means that a developer may legally retain authorization while a qualified external producer manufactures the product. Let $\tau_E(M)$, measured in C per project, denote the institutional barrier attached only to this retained entrusted route. The baseline representation is

$$\tau_E(0) = +\infty, \qquad \tau_E(1) = \bar{\tau}_E < +\infty, \qquad \bar{\tau}_E \geq 0. \tag{4}$$

There is no continuous implementation parameter in the baseline.

The route labels are I for internal manufacturing, E for retained entrusted manufacturing, T for transfer or another non-retained outside option, and A for abandonment or indefinite delay. If r_i denotes the project's route control, its common label domain is

$$r_i \in \{I, E, T, A\}. \tag{5}$$

Effective pre-MAH unavailability of E is encoded solely by $\tau_E(0) = +\infty$, which gives $W_i^E = -\infty$ once route values are assembled. Thus the common label domain is not a second policy channel. Under E, the developer remains the holder; entrusted production is not a transfer of ownership or authorization. The eventual optimizer is denoted by $r_i^*(q, m; M, p_m, \ell_i)$, but route values and the argmax are defined in the organizational-sorting section below.

2.4 Project-advancement and realization objects

Developer i chooses one common control $x_i \geq 0$, measured in X. Canonically, x_i is *original-drug innovation investment / project-advancement intensity*. It includes investment that advances viable original-drug projects toward a commercialization-relevant route-planning stage. It is broader than pure clinical-development effort, but it is not basic research, basic-compound discovery, patent-generating effort, or patent applications.

Research capability converts the control into the expected measure of viable planning-stage projects:

$$\lambda_i^{\mathrm{plan}} = a_i x_i. \tag{6}$$

The control x_i has units X, so $\lambda_i^{\mathrm{plan}}$ has units P.

Let p_m, measured in C/B, denote a conjectured price of qualified manufacturing service and let p_m^* denote its future market-consistent value. Let $\Omega_i(M, p_m)$, measured in C per planning-stage project, denote developer i's expected optimized project value. These objects are semantically reserved here: the sections below first build route values, then solve the advancement problem, and finally close the CMO fixed point.

Finally, let $s(q) \in [0, 1]$ be the exogenous, route-independent probability of downstream clinical or regulatory realization:

$$\Pr\{\text{downstream realization} \mid q\} = s(q). \tag{7}$$

The reform does not shift $s(q)$. If typed observed outcomes later require a class-specific probability $s_g(q)$, it is likewise exogenous and MAH-invariant.

2.5 Baseline timing

Stage 0. **Institutional regime.** The value of M is realized and publicly observed. It changes only the institutional barrier in (4); the infinite/finite wedge encodes the effective unavailability/availability of route E.

Stage 1. **Ex ante project advancement.** Developer i observes (a_i, k_i, ℓ_i, M), anticipates the market-consistent service price p_m^* and future route opportunities, and chooses x_i. Equation (6) determines the expected measure of viable projects reaching the route-planning stage.

Stage 2. **Project characteristics.** Each planning-stage project draws (q, m) according to (2). The optional label g records an empirical class but does not create another advancement choice.

Stage 3. **Commercialization organization.** After observing (q, m) and the manufacturing-service price, the developer checks each retained route's up-front financing requirement against ℓ_i, then chooses a route from the domain in (5). There is no borrowing or separate financing decision. Actual holder–producer separation is observed only if the realized route is E, which occurs after the Stage 1 choice.

Stage 4. **Downstream realization.** A retained project realizes downstream commercial value with the exogenous probability in (7). Planning-stage projects, route assignments, and realized products are distinct objects.

Stage 5. **Manufacturing-service consistency.** Aggregate entrusted-capacity demand and qualified-capacity supply must support the same p_m^* that developers anticipated. This is a simultaneous equilibrium consistency condition, not an unanticipated event occurring after Stage 4.

Upstream scientific research and patent generation may have occurred before Stage 0 and may help explain predetermined a_i. They are outside the baseline decision timing.

2.6 Mechanism and exclusion boundaries

The only direct reform arrow is

$$M \longrightarrow \tau_E(M) \quad \text{whose infinite/finite value encodes route } E \text{ availability.} \tag{8}$$

The intended ex ante channel, conditional on the value and optimization results derived below, is

$$M \longrightarrow \text{anticipated availability/value of } E \longrightarrow \Omega_i \longrightarrow x_i^*. \tag{9}$$

The later realization order is

$$\begin{aligned} x_i^* &\longrightarrow \text{planning-stage projects} \longrightarrow r_i^* \\ &\longrightarrow \text{observed holder–producer separation} \longrightarrow \text{realized products.} \end{aligned} \tag{10}$$

Equations (9) and (10) are timing and dependency statements, not unconditional comparative-static propositions.

In particular, M does not directly change a_i, k_i, ℓ_i, z_j, q, m, F, ρ_g, F_g, $s(q)$, or $s_g(q)$. It does not directly change route financing requirements and does not directly lower p_m^*. Nothing in this section implies an increase in patent applications, upstream scientific research, or one novelty class relative to the other.

The exact-category registry, dimensions, and forbidden interpretations are fixed by the definitions above. The equilibrium section below closes the deferred fixed point without adding another policy channel.

3 Demand and Operating Profit

This section derives the commercial value kernel used by later manufacturing routes. It conditions on a project value shifter $q > 0$ and a positive marginal manufacturing cost $c > 0$. The route technologies that generate c are defined in the next section. Product price p is distinct from the qualified manufacturing-service price p_m.

Let Y denote physical product-output units per operating period. Both p and c are measured in C/Y, demand is measured in Y per period, one-period operating profit in C per period, and present value in C per commercially active product.

3.1 Residual demand and the pricing problem

Residual demand for a successfully commercialized project is

$$y(p;q) = Aqp^{-\varepsilon}, \qquad A > 0, \quad q > 0, \quad \varepsilon > 1. \tag{11}$$

The scale A, project shifter q, and elasticity ε are exogenous to the institutional regime. Conditional on q and c, the developer chooses product price from $p \in [c, \infty)$:

$$\max_{p \geq c} \, (p - c)Aqp^{-\varepsilon}. \tag{12}$$

Holding A, q, c, ε fixed, the derivative with respect to the product price is

$$\begin{aligned} \frac{\partial}{\partial p}\left[(p-c)Aqp^{-\varepsilon}\right] &= Aqp^{-\varepsilon} - \varepsilon Aq(p-c)p^{-\varepsilon-1} \\ &= Aqp^{-\varepsilon-1}\left[\varepsilon c - (\varepsilon - 1)p\right]. \end{aligned} \tag{13}$$

Since $Aqp^{-\varepsilon-1} > 0$, the derivative has the sign of the bracketed affine term. That term crosses zero exactly once, giving

$$p^*(c) = \frac{\varepsilon}{\varepsilon - 1}c. \tag{14}$$

Because $\varepsilon > 1$, $p^*(c) > c$, so the solution is feasible and interior.

For an independent second-order check, differentiate (13). At the stationary point, the term multiplying the derivative of $p^{-\varepsilon-1}$ vanishes, leaving

$$\frac{\partial^2}{\partial p^2}\left[(p-c)Aqp^{-\varepsilon}\right]\bigg|_{p=p^*(c)} = -Aq(\varepsilon - 1)\left[p^*(c)\right]^{-\varepsilon-1} < 0. \tag{15}$$

The solution is also the unique global maximum. At the lower boundary $p = c$, profit is zero and the derivative is positive. The derivative is positive below $p^*(c)$ and negative above it, while profit converges to zero as p tends to infinity. Hence there is no competing boundary maximum.

3.2 Optimized one-period operating profit

Substituting (14) into the objective gives

$$\begin{aligned} \pi(q,c) &= \left(\frac{c}{\varepsilon - 1}\right) Aq \left(\frac{\varepsilon c}{\varepsilon - 1}\right)^{-\varepsilon} \\ &= Aq\frac{(\varepsilon - 1)^{\varepsilon - 1}}{\varepsilon^{\varepsilon}}c^{1-\varepsilon}. \end{aligned} \tag{16}$$

The units are consistent: the price–cost margin is in C/Y, demand is in Y per period, and their product is in C per period.

Holding A, ε, and the non-differentiated argument fixed,

$$\frac{\partial \pi(q,c)}{\partial q} = \frac{\pi(q,c)}{q} > 0, \qquad \frac{\partial \pi(q,c)}{\partial c} = (1-\varepsilon)\frac{\pi(q,c)}{c} < 0. \tag{17}$$

The cost elasticity of optimized operating profit is $1-\varepsilon$. These are continuous conditional derivatives in q and c, not a derivative with respect to the binary institutional regime.

3.3 Present value conditional on commercialization

Let $\beta \in (0,1)$ be the one-period discount factor. Conditional on an already commercialized product, let $\varphi \in [0,1)$ be the probability that it remains commercially active for one more period. The parameter φ is not the downstream realization probability $s(q)$: the latter applies before the operating stream begins and enters the route values defined below.

With a stationary conditional operating profit, the expected present value is the convergent geometric sum

$$\begin{aligned} R(q,c) &= \sum_{t=0}^{\infty} (\beta\varphi)^t \pi(q,c) \\ &= \frac{\pi(q,c)}{1-\beta\varphi}, \qquad 0 \le \beta\varphi < 1. \end{aligned} \tag{18}$$

Consequently, holding $\beta, \varphi, A, \varepsilon$ and the non-differentiated argument fixed,

$$\frac{\partial R(q,c)}{\partial q} = \frac{R(q,c)}{q} > 0, \qquad \frac{\partial R(q,c)}{\partial c} = (1-\varepsilon)\frac{R(q,c)}{c} < 0. \tag{19}$$

3.4 Boundary and accounting checks

The commercial-value kernel satisfies the following checks:

1. As $q \downarrow 0$, both $\pi(q,c)$ and $R(q,c)$ converge to zero.
2. As $c \to \infty$, both values converge to zero because $1-\varepsilon < 0$.
3. If $\varphi = 0$, the commercial stream lasts one period in expectation and $R(q,c) = \pi(q,c)$.
4. As $\varphi \uparrow 1$, $R(q,c) \to \pi(q,c)/(1-\beta)$, which remains finite because $\beta < 1$. The endpoint is a limiting check, not an admissible value under the frozen primitive domain $\varphi \in [0,1)$.
5. The restriction $\varepsilon > 1$ is necessary for a finite interior markup price under this residual-demand specification.

The object $R(q,c)$ is gross operating present value. The marginal cost c is subtracted exactly once through the operating margin $p-c$. Route-specific fixed organizational costs, the institutional burden on route E, and any separately defined qualified-capacity payment are absent from R; later route-value equations must specify whether each manufacturing cost is embedded in a route-specific marginal cost or subtracted separately, and must not do both. Reduced-form return shortcuts are inactive in the baseline; the model uses only the explicit chain $y \to \pi \to R$.

Nothing in this section gives MAH a direct effect on A, q, ε, β, or φ. Thus this section establishes a policy-invariant commercial value kernel, not a policy comparative static.

4 Internal and Entrusted Manufacturing Technologies

This section defines the manufacturing primitives that later enter route values. It does not construct those values, choose a route, or clear the qualified manufacturing-service market. All technology primitives are invariant to the MAH regime. The only direct institutional wedge remains $\tau_E(M)$.

4.1 Internal manufacturing

For a project with manufacturing requirement $m > 0$ and a developer with internal manufacturing capability $k_i > 0$, let

$$c_I(m, k_i) > 0, \qquad c_{I,m}(m, k_i) > 0, \qquad c_{I,k}(m, k_i) < 0. \tag{20}$$

The function c_I is technological marginal manufacturing cost, measured in C/Y. A more demanding project raises this cost, while greater internal capability lowers it on the feasible domain.

Internal production also requires a finite project-level production-readiness/setup cost $F_I(m, k_i)$, measured in C:

$$0 \le F_I(m, k_i) < +\infty, \qquad F_{I,m}(m, k_i) > 0, \qquad F_{I,k}(m, k_i) < 0. \tag{21}$$

Low internal capability therefore makes internalization expensive but does not make it technologically impossible by convention.

Let

$$J_I(m, k_i) > 0, \qquad J_{I,m}(m, k_i) > 0, \qquad J_{I,k}(m, k_i) < 0 \tag{22}$$

be the up-front financing commitment required to organize internal manufacturing. Route I is financially feasible exactly when $J_I(m, k_i) \le \ell_i$. The object J_I is a liquidity requirement, not an additional real cost. One possible accounting interpretation is $0 < J_I \le F_I$, but no result requires that inequality.

4.2 Qualified external production

Under entrusted manufacturing, a qualified external producer uses the technological marginal-cost kernel

$$c_E(m) > 0, \tag{23}$$

also measured in C/Y. This kernel does not depend on the developer's internal capability k_i: manufacturing is performed by the qualified external producer.

An entrusted project requires

$$b(m) > 0, \qquad b'(m) > 0, \tag{24}$$

units of qualified manufacturing-service capacity. The object $b(m)$ is a physical capacity requirement in B per project. It is not a route share, a supply curve, or an exogenous measure of scarcity relief.

The real technology-transfer, validation, and production-readiness cost is

$$F_E(m) \ge 0, \tag{25}$$

measured in C per project. In addition, the developer bears a residual holder-side responsibility and coordination burden

$$\mu_E \ge 0. \tag{26}$$

The burden μ_E is not eliminated by MAH.

Retained entrusted production also requires the up-front financing commitment

$$J_E(m) > 0, \qquad J_E'(m) > 0. \tag{27}$$

Route E is financially feasible exactly when $J_E(m) \le \ell_i$, in addition to being institutionally available. The commitment covers the liquidity needed for transfer, validation, initial contracting, and holder-side preparation. It is not subtracted from route value.

The accounting roles are distinct. The technological marginal cost $c_E(m)$ is the input to the operating-value kernel derived above. The separate monetary capacity payment $p_m b(m)$, the readiness cost $F_E(m)$, the holder burden μ_E, and the institutional barrier $\tau_E(M)$ are not embedded in $c_E(m)$. Each can therefore enter a later route value at most once. Likewise, J_I and J_E are feasibility requirements and are not counted again as payoff costs.

4.3 Organizational and institutional distinction

Routes I and E are technologically and organizationally distinct:

- under I, the developer remains the holder and manufactures using its own capability k_i;
- under E, the developer remains the authorization holder, retains holder responsibility, and a qualified external producer manufactures;
- route E is not the transfer or out-licensing route T, under which the retained holder–producer organization is not used.

The only direct policy object is

$$M \in \{0,1\}, \qquad \tau_E(0) = +\infty, \qquad \tau_E(1) = \bar{\tau}_E < +\infty. \tag{28}$$

The regime does not change c_I, F_I, J_I, c_E, b, F_E, J_E, μ_E, a_i, k_i, ℓ_i, q, m, $s(q)$, or $s_g(q)$. In particular, this technology block does not posit a fall in p_m^* or an outward shift of CMO supply. Price and capacity are equilibrium objects determined in the qualified manufacturing-service market below.

4.4 Boundary checks

Increasing k_i lowers internal marginal cost, setup cost, and the financing commitment J_I, while increasing m raises all three. A sufficiently well-financed low-k_i developer can still internalize. In the limiting approximation $J_I(m, k_i) \to +\infty$ for very low k_i, the old hard-infeasibility convention is recovered. The entrusted technology does not use k_i, but a larger m raises both its qualified-capacity requirement $b(m)$ and financing commitment $J_E(m)$. None of these statements is a route-choice result: route values, financing-adjusted values, outside options, and deterministic sorting are defined in the next section.

5 Route Values and Organizational Sorting

This section composes the commercial-value kernel with the internal and entrusted manufacturing technologies defined above. It treats the qualified manufacturing-service price p_m as fixed and conjectured. Market consistency for p_m^* is imposed in the capacity-market section below.

5.1 Route values

The economic value of internal production is

$$W_i^I(q, m) = s(q)R(q, c_I(m, k_i)) - F_I(m, k_i). \tag{29}$$

Its financing-adjusted value is

$$\widetilde{W}_i^I(q, m; \ell_i) = \begin{cases} W_i^I(q, m), & J_I(m, k_i) \le \ell_i, \\ -\infty, & J_I(m, k_i) > \ell_i. \end{cases} \tag{30}$$

At a conjectured CMO capacity price p_m, retained entrusted manufacturing has value

$$\begin{aligned} W_i^E(q, m; M, p_m) = s(q)R(q, c_E(m)) - F_E(m) - p_m b(m) \\ - \mu_E - \tau_E(M). \end{aligned} \tag{31}$$

The developer remains the authorization holder under E; the qualified external producer manufactures. Each route cost is subtracted exactly once. In particular, $p_m b(m)$ is not contained in $c_E(m)$. The financing-adjusted entrusted value is

$$\widetilde{W}_i^E(q,m;M,p_m,\ell_i)=\begin{cases}W_i^E(q,m;M,p_m), & J_E(m)\le \ell_i,\\ -\infty, & J_E(m)>\ell_i.\end{cases} \tag{32}$$

Neither J_I nor J_E is subtracted from route value: each is a liquidity threshold, not a second copy of a real cost.

Let the noncore transfer/out-license outside value be a finite continuous function $T(q,m)$, and normalize abandonment or indefinite delay to zero:

$$W^T(q,m)=T(q,m), \qquad W^A=0. \tag{33}$$

No separate transfer market is added.

5.2 Deterministic route choice

The optimized value of a planning-stage project is

$$\widetilde{W}_i(q,m;M,p_m,\ell_i)=\max\left\{\widetilde{W}_i^I(q,m;\ell_i),\widetilde{W}_i^E(q,m;M,p_m,\ell_i),W^T(q,m),W^A\right\}. \tag{34}$$

The route choice is

$$r_i^*(q,m;M,p_m,\ell_i)=\arg\max_{r\in\{I,E,T,A\}}\begin{cases}\widetilde{W}_i^I(q,m;\ell_i), & r=I,\\ \widetilde{W}_i^E(q,m;M,p_m,\ell_i), & r=E,\\ W^T(q,m), & r=T,\\ W^A, & r=A.\end{cases} \tag{35}$$

Continuous project and developer heterogeneity makes exact ties a measure-zero event, so the choice is single valued almost surely. This is a deterministic comparison; it generates no probabilistic route share.

When $M=0$, $\tau_E(0)=+\infty$ implies $\widetilde{W}_i^E=-\infty$. At fixed p_m, the binary reform comparison is therefore

$$\begin{aligned}W_i^0(q,m;\ell_i)&=\max\{\widetilde{W}_i^I(q,m;\ell_i),W^T(q,m),0\},\\ W_i^1(q,m;p_m,\ell_i)&=\max\{W_i^0(q,m;\ell_i),\widetilde{W}_i^E(q,m;1,p_m,\ell_i)\},\\ W_i^1-W_i^0&=[\widetilde{W}_i^E-W_i^0]_+\ge 0.\end{aligned} \tag{36}$$

The gain is exactly zero, and MAH has zero project-value and route-choice effect, whenever the newly feasible entrusted value fails to exceed the best pre-reform route.

5.3 Internal–entrusted value gap

For fixed q,m,M,p_m, define

$$\begin{aligned}\Delta_{IE}(k_i;q,m,M,p_m)&=W_i^I(q,m)-W_i^E(q,m;M,p_m)\\ &=s(q)[R(q,c_I(m,k_i))-R(q,c_E(m))]-F_I(m,k_i)\\ &\quad+F_E(m)+p_m b(m)+\mu_E+\tau_E(M).\end{aligned} \tag{37}$$

Conditional on both retained routes being financially feasible, only the internal technology terms in the economic value gap vary with k_i. Holding q,m,M,p_m fixed,

$$\frac{\partial\Delta_{IE}}{\partial k_i}=s(q)R_c(q,c_I(m,k_i))\,c_{I,k}(m,k_i)-F_{I,k}(m,k_i)>0. \tag{38}$$

The first term is weakly positive because $s(q)\ge 0$, $R_c<0$, and $c_{I,k}<0$; the second is strictly positive because $F_{I,k}<0$. Hence the entire derivative is strictly positive, including the boundary case $s(q)=0$.

5.4 Conditional organizational cutoff

Consider a finite entrusted barrier and fix ℓ_i such that route E is financeable. Since $J_{I,k} < 0$, the set of capabilities at which route I is also financeable is an upper interval. On a connected portion of that common-financeability set, suppose the continuous value gap satisfies the endpoint crossing conditions

$$\Delta_{IE}(k_L) < 0, \qquad \Delta_{IE}(k_H) > 0, \qquad k_L < k_H.$$

Continuity gives existence of a root, and (38) gives strict monotonicity and therefore uniqueness. Denote the unique root by

$$\Delta_{IE}(k^*(q, m; p_m, M); q, m, M, p_m) = 0. \tag{39}$$

Conditional on both routes being financeable and both dominating transfer and abandonment,

$$k_i < k^* \Rightarrow r_i^* = E, \qquad k_i > k^* \Rightarrow r_i^* = I. \tag{40}$$

If T or A dominates, the I/E cutoff does not determine the realized route.

For a finite wedge, let the arguments τ_E and p_m vary locally while holding q, m fixed. Since

$$\frac{\partial \Delta_{IE}}{\partial \tau_E} = 1, \qquad \frac{\partial \Delta_{IE}}{\partial p_m} = b(m) > 0,$$

the implicit-function theorem yields

$$\frac{\partial k^*}{\partial \tau_E} = -\frac{1}{\Delta_{IE,k}(k^*)} < 0, \qquad \frac{\partial k^*}{\partial p_m} = -\frac{b(m)}{\Delta_{IE,k}(k^*)} < 0. \tag{41}$$

A lower finite institutional barrier raises k^*, expanding the entrusted-production region $k_i < k^*$. A higher fixed CMO price lowers k^*, contracting that region. The second derivative is a fixed-price comparative static, not an equilibrium derivative through p_m^*.

5.5 Financing corridor and MAH-relevant set

The financing-adjusted MAH-relevant set is

$$\mathcal{C}_i(p_m, \ell_i) = \left\{(q, m) : J_E(m) \le \ell_i, \quad W_i^E(q, m; 1, p_m) > W_i^0(q, m; \ell_i)\right\}. \tag{42}$$

The fixed-price gain is strict exactly on this set.

For a fixed (q, m, k_i, p_m), suppose the locally relevant capital ordering

$$J_E(m) < J_I(m, k_i) \tag{43}$$

holds. This is a proposition-specific condition for research-oriented or low-internal-capability developers, not a global assumption. Financing capacity then creates three regions:

1. $\ell_i < J_E(m)$: neither retained route is financeable and MAH has no project-value effect;
2. $J_E(m) \le \ell_i < J_I(m, k_i)$: entrusted production is financeable but internalization is not. The MAH gain is strict if and only if $W_i^E > \max\{W^T, 0\}$;
3. $\ell_i \ge J_I(m, k_i)$: both retained routes are financeable and the gain is $[W_i^E - \max\{W_i^I, W^T, 0\}]_+$.

The treatment effect can therefore rise at the lower corridor boundary and fall, remain positive, or become zero when internalization returns to the old choice set. It has no global monotonic sign in ℓ_i. By contrast, within either fixed regime, a larger ℓ_i weakly expands the feasible choice set and weakly raises optimized project value.

5.6 Organizational sorting result

Suppose the commercial-value and technology restrictions above hold, both retained routes are financeable, the entrusted barrier is finite, the endpoint crossing conditions hold on the common-financeability interval, and ties have measure zero. Then the internal-minus-entrusted economic value gap is strictly increasing in developer manufacturing capability, a unique finite I/E cutoff exists, and the conditional sorting and implicit derivatives in (40)–(41) follow. The proposition is conditional: it does not displace the valid transfer and abandonment outside options.

5.7 Zero-effect and non-tautology boundary

The route comparison is not tautological. Internal value and its financing threshold depend on developer capability through c_I and F_I; entrusted value instead uses qualified-external technology, a physical capacity requirement, a retained holder burden, its own financing threshold, and the institutional wedge. MAH can nevertheless have zero effect when E is not financeable, remains below the pre-reform maximum, when T or A dominates, or when the relevant project–firm pair lies outside the finite cutoff-crossing region. No smooth random-utility term, inclusive-value object, borrowing decision, or probabilistic route share is used.

6 Project Development and Advancement

This section solves the ex ante advancement decision at a fixed conjectured qualified manufacturing-service price p_m. It uses the deterministic optimized route value defined above. CMO market consistency and the feedback that selects p_m^* are imposed in the capacity-market section below.

6.1 Innovation-object boundary and planned-project technology

Developer i chooses one common control $x_i \geq 0$. Its canonical meaning is original-drug innovation investment / project-advancement intensity. It is broader than pure clinical-development effort, but it is not patent applications, patent-generating effort, basic research, or upstream scientific discovery. There is no class-specific control x_{ig}.

Predetermined project-advancement productivity $a_i > 0$ converts the control into the expected measure of viable projects reaching route planning:

$$\lambda_i^{\text{plan}} = a_i x_i. \tag{44}$$

The control has units X, and a_i has units P/X, so λ_i^{plan} has units P.

The advancement-cost function is

$$C_X(x_i) = \frac{\kappa}{1+\nu} x_i^{1+\nu}, \qquad \kappa > 0, \quad \nu > 0. \tag{45}$$

Here κ has units $\mathsf{C}/\mathsf{X}^{1+\nu}$, so C_X is measured in currency per developer and decision cohort. Its marginal cost is $C_X'(x_i) = \kappa x_i^{\nu}$. The boundary $\nu = 1$ gives a quadratic advancement cost.

6.2 Expected optimized project value

For developer i, institutional regime M, and fixed conjectured price p_m, define

$$\begin{aligned}\Omega_i(M, p_m) &= \mathbb{E}_F\left[\widetilde{W}_i(q, m; M, p_m, \ell_i)\right] \\ &= \int \widetilde{W}_i(q, m; M, p_m, \ell_i)\, dF(q, m).\end{aligned} \tag{46}$$

The subscript i includes the dependence of route value on developer characteristics, especially manufacturing capability k_i and financing capacity ℓ_i. Because the route-value maximum contains abandonment with value zero, $\widetilde{W}_i(q, m; M, p_m, \ell_i) \geq 0$. Under the stated integrability condition,

$$0 \leq \Omega_i(M, p_m) < +\infty.$$

This expectation replaces the old theoretical-baseline inclusive-value object; no random-utility aggregation is used.

6.3 Ex ante advancement problem

The developer solves

$$\max_{x_i \geq 0} \left\{ \beta a_i x_i \Omega_i(M, p_m) - \frac{\kappa}{1+\nu} x_i^{1+\nu} \right\}. \tag{47}$$

The decision precedes the (q, m) draw and downstream route choice. Conditional on M, p_m, a_i, k_i, ℓ_i, the expected value $\Omega_i(M, p_m)$ is therefore fixed with respect to the individual choice x_i.

The necessary and sufficient Kuhn–Tucker conditions are

$$\begin{gathered} x_i \geq 0, \qquad \kappa x_i^{\nu} - \beta a_i \Omega_i(M, p_m) \geq 0, \\ x_i \left[\kappa x_i^{\nu} - \beta a_i \Omega_i(M, p_m) \right] = 0. \end{gathered} \tag{48}$$

For an interior solution, the FOC is

$$\kappa x_i^{\nu} = \beta a_i \Omega_i(M, p_m). \tag{49}$$

For $x_i > 0$, the objective's second derivative is

$$-\kappa \nu x_i^{\nu-1} < 0. \tag{50}$$

More generally, $x_i^{1+\nu}$ is strictly convex on the nonnegative domain for every $\nu > 0$. Hence the objective is strictly concave. Its superlinear cost dominates the linear benefit as $x_i \to \infty$, so a unique maximizer exists.

The unique optimal common advancement intensity is

$$x_i^*(M, p_m) = \left[\frac{\beta a_i}{\kappa} \Omega_i(M, p_m) \right]^{1/\nu}. \tag{51}$$

If $\Omega_i = 0$, the formula gives the unique corner $x_i^* = 0$. If $\Omega_i > 0$, the solution is interior. Dimensionally, $\beta a_i \Omega_i / \kappa$ has units X^{ν}, so the solution has units X.

6.4 Fixed-price binary MAH channel

At a fixed p_m, deterministic route choice implies the pointwise weak ordering

$$\widetilde{W}_i(q, m; 1, p_m, \ell_i) \geq \widetilde{W}_i(q, m; 0, p_m, \ell_i).$$

Taking expectations gives the finite policy comparison

$$\Omega_i(1, p_m) - \Omega_i(0, p_m) = \int \left[\widetilde{W}_i(q, m; 1, p_m, \ell_i) - \widetilde{W}_i(q, m; 0, p_m, \ell_i) \right] dF(q, m) \geq 0. \tag{52}$$

For positive Ω_i,

$$\frac{\partial x_i^*}{\partial \Omega_i} = \frac{1}{\nu} \left(\frac{\beta a_i}{\kappa} \right)^{1/\nu} \Omega_i^{1/\nu - 1} > 0.$$

Thus, at fixed p_m,

$$\Omega_i(1, p_m) > \Omega_i(0, p_m) \quad \Longleftrightarrow \quad x_i^*(1, p_m) > x_i^*(0, p_m).$$

The strict inequality in expected value requires a positive-probability set of projects in $\mathcal{C}_i(p_m, \ell_i)$, where entrusted production is financeable and strictly improves the pre-reform maximum. Otherwise both expected value and advancement can have a zero reform effect.

This is a binary finite comparison, not a derivative with respect to M. It is also a fixed-price result, not the equilibrium-price comparative static through p_m^*. The regime changes neither a_i, κ, ν, the cost function, nor the project distribution. MAH affects x_i^* only through Ω_i. Manufacturing capability k_i affects Ω_i through route values and sorting, not through $a_i x_i$ or $C_X(x_i)$. Financing capacity also affects advancement only through the set of financeable commercialization routes. It does not enter the advancement FOC directly.

Within a fixed regime, higher ℓ_i weakly expands the feasible route set, so Ω_i and x_i^* are weakly increasing in the finite-comparison sense. The binary reform response $\Delta x_i(\ell_i) = x_i^*(1, p_m, \ell_i) - x_i^*(0, p_m, \ell_i)$ has no global monotonic sign in ℓ_i: financing first makes E feasible and may later restore I as an old option. Derivatives at the financing thresholds need not exist.

6.5 Downstream project-value gap

For a useful value decomposition, let B_i denote the continuation-value benchmark before one additional viable project reaches route planning, and let K_i denote value after it reaches that stage. Define

$$K_i = B_i + \Omega_i, \qquad K_i - B_i = \Omega_i. \tag{53}$$

Substituting this accounting identity into the interior FOC gives

$$C_X'(x_i) = \beta a_i (K_i - B_i). \tag{54}$$

This is an Akcigit-style downstream project-value interpretation. The auxiliaries B_i and K_i are not recursive state variables, and the identity does not claim that MAH directly raises scientific-research productivity.

6.6 Scope boundary

The baseline contains one common x_i, one cost function C_X, one expected optimized project value Ω_i, and predetermined commercialization financing capacity ℓ_i. It contains no separate research and development controls, no financing constraint on x_i, no borrowing choice, no finance market, no patent outcome, no novelty-class-specific control, and no CMO market-clearing condition in this block. Project advancement is ex ante; financing feasibility, route choice, and observed holder–producer separation are downstream.

7 Qualified Manufacturing-Service Market

This section closes only the market for qualified manufacturing-service capacity. Developers and qualified suppliers take a conjectured price p_m as given when making their individual choices. The market-consistent price p_m^* is then selected by one scalar clearing equation. No entry, labor, capital, product-market, or welfare market is added.

7.1 Qualified-capacity supply

Qualified supplier j, with predetermined efficiency z_j, chooses capacity $s_j \geq 0$ to solve

$$\max_{s_j \geq 0} \left\{ p_m s_j - \Psi(s_j; z_j) \right\}. \tag{55}$$

Capacity has units B per decision cohort, p_m has units C/B, and Ψ has units C.

For every z_j, assume

$$\Psi(0; z_j) = 0, \qquad \Psi_s(0; z_j) = 0,$$

$$\Psi_s > 0 \text{ for } s_j > 0, \qquad \Psi_{ss} > 0, \qquad \Psi_{sz} < 0,$$

and $\Psi_s(s_j; z_j) \to +\infty$ as $s_j \to +\infty$. At $p_m = 0$, the unique optimum is $s_j^* = 0$. For $p_m > 0$, the solution is interior and satisfies

$$p_m = \Psi_s\big(s_j^*(p_m, z_j); z_j\big)\,. \tag{56}$$

The objective's second derivative is $-\Psi_{ss} < 0$, so the capacity choice is unique. The implicit-function theorem gives

$$\frac{\partial s_j^*}{\partial p_m} = \frac{1}{\Psi_{ss}} > 0, \qquad \frac{\partial s_j^*}{\partial z_j} = -\frac{\Psi_{sz}}{\Psi_{ss}} > 0. \tag{57}$$

Aggregate qualified-capacity supply is

$$S_m(p_m) = \int s_j^*(p_m, z)\, dH_C(z). \tag{58}$$

Under the stated integrability conditions, S_m is continuous and strictly increasing, $S_m(0) = 0$, and $S_m(p_m) \to +\infty$ as $p_m \to +\infty$. Supply technology and H_C are invariant to M; MAH creates no direct supply shift.

7.2 Entrusted-capacity demand per planning-stage project

At a candidate price p_m, define developer i's expected qualified capacity per planning-stage project as

$$\chi_i^E(p_m; M, \ell_i) = \int b(m)\mathbf{1}\left\{r_i^*(q, m; M, p_m, \ell_i) = E\right\} dF(q, m). \tag{59}$$

This is an integral of deterministic route choices, not a probabilistic route-share formula.

The entrusted value falls with price at slope $-b(m) < 0$, while the other route values do not contain p_m. Consequently, for each project the set of prices at which E is chosen is an interval from below: once a higher price makes E lose to another route, a still higher price cannot restore it. Hence $\chi_i^E(p_m; M, \ell_i)$ is weakly decreasing in p_m. The financing indicator $\mathbf{1}\{J_E(m) \le \ell_i\}$ does not depend on price.

Away from the measure-zero set of route ties, the envelope derivative of the optimized project value is

$$\frac{\partial \widetilde{W}_i(q, m; M, p_m, \ell_i)}{\partial p_m} = -b(m)\mathbf{1}\{r_i^*(q, m; M, p_m, \ell_i) = E\}.$$

Dominated convergence then gives

$$\frac{\partial \Omega_i(M, p_m)}{\partial p_m} = -\chi_i^E(p_m; M, \ell_i) \le 0 \quad \text{almost everywhere in } p_m. \tag{60}$$

For $\Omega_i > 0$, the advancement optimizer therefore satisfies

$$\frac{\partial x_i^*(M, p_m)}{\partial p_m} = -\frac{x_i^*(M, p_m)}{\nu \Omega_i(M, p_m)} \chi_i^E(p_m; M, \ell_i) \le 0. \tag{61}$$

At the zero-value corner, $x_i^* = 0$; globally the optimal advancement intensity remains weakly decreasing in p_m.

7.3 Aggregate demand

Developer i's expected qualified-capacity demand is

$$a_i x_i^*(M, p_m)\chi_i^E(p_m; M, \ell_i).$$

Aggregating over developer heterogeneity gives study-related demand

$$D_m^{\mathrm{MAH}}(p_m; M) = \int a_i x_i^*(M, p_m)\chi_i^E(p_m; M, \ell_i)dH(a, k, \ell). \tag{62}$$

This expression contains both price feedbacks required by the model: x_i^* responds through expected project value, and χ_i^E responds through deterministic route selection.

Both factors are nonnegative and weakly decreasing in p_m, so D_m^{MAH} is weakly decreasing. Individual route indicators can jump at a cutoff. Aggregate demand is nevertheless continuous: when $p_m^n \to p_m$, route indicators converge for every project–developer pair except the zero-measure route-tie set, x_i^* is continuous through Ω_i, and the common integrable envelope permits dominated convergence. The joint distribution also assigns zero mass to the financing thresholds $\ell_i = J_I(m, k_i)$ and $\ell_i = J_E(m)$. Thus aggregate regularity comes from continuous heterogeneity, not smooth random utility.

Let $D_m^B(p_m)$ be finite, continuous, nonnegative, weakly decreasing background demand with

$$D_m^B(0) > 0, \qquad \lim_{p_m \to \infty} D_m^B(p_m) = 0.$$

It is policy invariant and allows a CMO market to exist before MAH. Total demand is

$$D_m(p_m; M) = D_m^B(p_m) + D_m^{\mathrm{MAH}}(p_m; M). \tag{63}$$

It is finite, continuous, nonnegative, and weakly decreasing.

7.4 Market clearing, existence, and uniqueness

The qualified-capacity market clears at

$$D_m(p_m^*; M) = S_m(p_m^*). \tag{64}$$

At zero price,

$$D_m(0; M) \geq D_m^B(0) > 0 = S_m(0). \tag{65}$$

As $p_m \to \infty$, $W_i^E \to -\infty$ pointwise because $b(m) > 0$, so $\chi_i^E \to 0$ and study-related demand tends to zero under the integrable envelope. Background demand also tends to zero, whereas aggregate supply tends to infinity. Continuity therefore gives at least one positive clearing price.

Total demand is weakly decreasing and aggregate supply is strictly increasing. Their difference is strictly decreasing, so the clearing price is unique.

7.5 Closed scalar solution order

For each regime M, the equilibrium can be computed without an unresolved cycle:

1. conjecture p_m;
2. evaluate financing feasibility, route values, and deterministic $r_i^*(q, m; M, p_m, \ell_i)$;
3. integrate $\Omega_i(M, p_m)$ and compute $x_i^*(M, p_m)$;
4. integrate χ_i^E and D_m^{MAH};
5. solve supplier capacity and aggregate $S_m(p_m)$;
6. find the unique zero of $D_m(p_m; M) - S_m(p_m)$.

Every object on the right side is a function of the candidate price, leaving one scalar equation in p_m.

When $M = 0$, route E is unavailable, so $D_m^{\mathrm{MAH}}(p_m; 0) = 0$; background demand still supports the market. When $M = 1$, study demand is nonnegative and supply is unchanged. Hence the post-MAH clearing price is weakly above the pre-MAH price, and it is strictly higher if study demand is positive at the pre-MAH price. This is an endogenous scarcity-price response, not a direct policy shift in p_m^* or CMO technology.

7.6 Scope boundary

This section closes qualified manufacturing-service capacity only. It introduces no supplier entry, no direct policy supply shift, no logit route smoothing, and no additional market-clearing condition. The price response is generated jointly by supplier optimization, deterministic route selection, project advancement, and scalar market clearing.

8 Baseline MAH Partial Equilibrium

Fix the institutional regime $M \in \{0,1\}$, the developer distribution $H(a,k,\ell)$, the project distribution $F(q,m)$, the qualified-supplier distribution $H_C(z)$, and all other exogenous primitives introduced above. A baseline MAH partial equilibrium is the collection

$$\boxed{\left\{p_m^*,\ x_i^*,\ r_i^*(q,m),\ s_j^*\right\}_{i,j}}. \tag{66}$$

The suppressed arguments are $x_i^* = x_i^*(M, p_m^*)$, $r_i^*(q,m) = r_i^*(q,m;M,p_m^*,\ell_i)$, and $s_j^* = s_j^*(p_m^*, z_j)$. The collection contains exactly these four families of endogenous optimal objects.

It satisfies the following four conditions.

1. Optimal commercialization route. For every planning-stage project–developer pair outside the zero-measure tie set,

$$r_i^*(q,m;M,p_m^*,\ell_i) \in \arg \max_{r\in\{I,E,T,A\}} \left\{\widetilde{W}_i^I(q,m;\ell_i), \widetilde{W}_i^E(q,m;M,p_m^*,\ell_i), W^T(q,m), W^A\right\}, \tag{67}$$

with route E unavailable when $M = 0$. This is the deterministic route choice defined above; route E retains holder rights and is not route T. There is no financing choice: ℓ_i is predetermined and only screens the two retained routes against J_I and J_E.

2. Optimal project advancement. For every developer i, the common project-advancement control is

$$x_i^*(M,p_m^*) = \left[\frac{\beta a_i \Omega_i(M,p_m^*)}{\kappa}\right]^{1/\nu}, \tag{68}$$

including the zero-value corner $x_i^* = 0$. There is no class-specific x_{ig} control.

3. Optimal qualified-supplier capacity. For every qualified supplier j,

$$s_j^*(p_m^*, z_j) \in \arg\max_{s_j\geq 0} \left\{p_m^* s_j - \Psi(s_j;z_j)\right\}. \tag{69}$$

Strict convexity of Ψ makes this capacity choice unique. Supplier entry is not a margin of the model.

4. Qualified-capacity market clearing. The price p_m^* is the unique solution to

$$D_m(p_m^*;M) = S_m(p_m^*). \tag{70}$$

The qualified-capacity existence and uniqueness conditions apply. The demand and supply functions determine the price but are not additional members of the equilibrium collection.

8.1 Closure boundary

Equations (67)–(70) close only commercialization route choice, common project advancement, qualified-supplier capacity, and the manufacturing-service price. There is no entry condition and no labor-market, finance-market, capital-market, product-market aggregate, firm-distribution, or welfare-clearing condition. Product price $p^*(c)$ remains a conditional firm-level optimum from the pricing problem, not a second market-clearing price.

9 Required Baseline Propositions

This section states exactly six substantive propositions. The novelty result is a corollary of the MAH-relevant-project-set identity, not a seventh substantive proposition. All policy comparisons are finite comparisons between $M = 0$ and $M = 1$; the binary regime is never differentiated.

9.1 Proposition 1: Organizational sorting

Assumptions and fixed objects. Fix a project (q, m), a finite institutional wedge, regime $M = 1$, and a conjectured support price $p_m < \infty$. Fix financing capacity ℓ_i and hold all primitives except internal manufacturing capability k_i fixed. Use the technology and financing-requirement signs and the endpoint-crossing conditions stated above.

Claim. Internal production is financeable exactly when $J_I(m, k_i) \leq \ell_i$; entrusted production is financeable exactly when $J_E(m) \leq \ell_i$, in addition to being institutionally available. Since $J_{I,k} < 0$, stronger internal capability expands the internal-financeability set. Conditional on both retained routes being financeable, the economic internal-minus-entrusted value gap is strictly increasing in k_i:

$$\frac{\partial \Delta_{IE}(k_i; q, m, M, p_m)}{\partial k_i} = s(q) R_c(q, c_I) c_{I,k}(m, k_i) - F_{I,k}(m, k_i) > 0. \tag{71}$$

If the gap crosses from negative to positive on a connected interval where both retained routes are financeable, there is a unique $k^*(q, m; p_m, M)$. Conditional on both I and E dominating T and A, developers below the cutoff use E and developers above it use I.

Proof. The feasibility statements follow from the definitions of $\widetilde{W}_i^I$ and $\widetilde{W}_i^E$. The sign $J_{I,k} < 0$ implies that if internalization is financeable at k_i, it remains financeable at any higher capability holding (m, ℓ_i) fixed. Conditional on both routes being financeable, the commercial-value kernel gives $R_c < 0$, while the internal technology gives $c_{I,k} < 0$ and $F_{I,k} < 0$. The operating-cost term on the right side of (71) is weakly positive (and is zero when $s(q) = 0$), while the setup-cost term is strictly positive. Their sum is therefore strictly positive. Continuity and endpoint crossing give existence by the intermediate value theorem; strict monotonicity gives uniqueness. The sign of the gap on either side of its unique zero gives the conditional sorting statement. The comparison with T/A is required because an I/E ranking alone does not establish the globally selected route.

Sufficient conditions and zero-effect cases. The endpoint-crossing conditions are sufficient, not primitive conclusions. If they fail, a finite interior cutoff need not exist. If $J_I(m, k_i) > \ell_i$, internal production is financially infeasible even though its technological payoff remains finite. A low-k_i developer with sufficiently high ℓ_i can still internalize. As $k_i \to \infty$, the maintained technology signs weakly favor internal production and reduce its financing requirement. At $M = 0$, route E is unavailable and there is no finite pre-MAH I/E cutoff.

Economic interpretation. Internal capability reduces internal operating cost, setup burden, and the amount of up-front commitment that must be financed. Entrusted production is most useful to developers for which those internal advantages are weak, but it is selected only when it also beats transfer or delay. The result is organizational sorting, not a permanent firm-type taxonomy.

9.2 Proposition 2: The MAH-relevant project set

Assumptions and fixed objects. Fix developer i, its predetermined financing capacity ℓ_i, project draw (q, m), and support price p_m. The old route set is I, T, A; post-MAH adds E without removing an old option. Route values retain the definitions given above.

Definitions and claim. Define

$$W_i^0(q, m; \ell_i) = \max\{\widetilde{W}_i^I(q, m; \ell_i), W^T(q, m), 0\}, \tag{72}$$

$$W_i^1(q, m; p_m, \ell_i) = \max\{W_i^0(q, m; \ell_i), \widetilde{W}_i^E(q, m; 1, p_m, \ell_i)\}. \tag{73}$$

Then

$$W_i^1(q, m; p_m, \ell_i) - W_i^0(q, m; \ell_i) = \left[\widetilde{W}_i^E(q, m; 1, p_m, \ell_i) - W_i^0(q, m; \ell_i)\right]_+ . \tag{74}$$

The MAH-relevant project set is

$$\mathcal{C}_i(p_m, \ell_i) = \left\{(q, m) : J_E(m) \le \ell_i, \quad W_i^E(q, m; 1, p_m) > W_i^0(q, m; \ell_i)\right\}. \tag{75}$$

Only project draws in this set receive a strict direct value gain at the fixed support price.

Proof. For any real numbers a, b, $\max\{a, b\} - a = [b - a]_+$. Apply the identity with $a = W_i^0$ and $b = \widetilde{W}_i^E$. The gain is strictly positive exactly where the argument of the positive-part operator is positive, which is precisely $\mathcal{C}_i(p_m, \ell_i)$.

Sufficient conditions and zero-effect cases. A project receives a strict gain if and only if entrusted production is financeable and its value strictly exceeds every financeable old route at the support price. If $J_E > \ell_i$ or $W_i^E \le W_i^0$, the project gain is zero. If one of these conditions holds for every draw, $\mathcal{C}_i(p_m, \ell_i)$ is empty up to null sets. As $p_m \to \infty$, $W_i^E \to -\infty$, so the old value is preserved and the relevant set becomes empty.

Economic interpretation. MAH matters only where retained entrusted manufacturing is financeable and beats every option already available. Suppose locally that $J_E(m) < J_I(m, k_i)$. If $\ell_i < J_E$, neither retained route is financeable and the effect is zero. If $J_E \le \ell_i < J_I$, entrusted production is the only financeable retained route and yields a strict gain when it beats T/A. If $\ell_i \ge J_I$, internalization returns as an old option and the gain is $[W_i^E - \max\{W_i^I, W^T, 0\}]_+$. The reform response can therefore be strongest at intermediate financing capacity and is not globally monotone in ℓ_i.

9.3 Proposition 3: Project advancement and firm heterogeneity

Assumptions and fixed objects. Fix a support price p_m, financing capacity ℓ_i, policy-invariant primitives, and developer characteristics except for the characteristic varied in each comparison. Route values do not depend on research capability a_i. For the manufacturing-capability result, additionally impose the proposition-specific sufficient condition $\nu \ge 1$.

Proof: expected value and advancement response. For $h \in \{0, 1\}$, define

$$\Omega_i^h(p_m, \ell_i) = \int W_i^h(q, m; p_m, \ell_i)\, dF(q, m), \tag{76}$$

with the irrelevant p_m argument suppressed for $h = 0$. Proposition 2 implies

$$\Delta\Omega_i(p_m) \equiv \Omega_i^1(p_m) - \Omega_i^0 = \int \Big[\widetilde{W}_i^E(q,m;1,p_m,\ell_i) - W_i^0(q,m;\ell_i)\Big]_+ dF(q,m) \geq 0. \tag{77}$$

The common advancement response is

$$\Delta x_i(p_m) = \left(\frac{\beta a_i}{\kappa}\right)^{1/\nu} \Big[\big(\Omega_i^0 + \Delta\Omega_i(p_m)\big)^{1/\nu} - \big(\Omega_i^0\big)^{1/\nu}\Big]. \tag{78}$$

Because $t^{1/\nu}$ is strictly increasing for every $\nu > 0$,

$$\Delta x_i(p_m) > 0 \quad \Longleftrightarrow \quad \Delta\Omega_i(p_m) > 0.$$

Research-capability scaling. Holding $k_i, p_m, \Omega_i^0, \Delta\Omega_i$ and all primitives fixed, a positive response satisfies

$$\frac{\partial \Delta x_i(p_m)}{\partial a_i} = \frac{\Delta x_i(p_m)}{\nu a_i} > 0. \tag{79}$$

Thus the level response is stronger at higher a_i whenever the MAH-relevant set has positive expected surplus.

Capability result. The internal value is nondecreasing in k_i, whereas the other old route values and entrusted value are independent of k_i. Therefore $\Omega_i^0(k_i)$ is nondecreasing and $\Delta\Omega_i(k_i; p_m)$ is nonincreasing. When $\nu \geq 1$, $h(t) = t^{1/\nu}$ is increasing and concave, so $h(u + d) - h(u)$ is nondecreasing in d and nonincreasing in u. Applying this result to $u = \Omega_i^0(k_i)$ and $d = \Delta\Omega_i(k_i; p_m)$ yields

$$k_i' \geq k_i \quad \Longrightarrow \quad \Delta x_i(k_i'; p_m) \leq \Delta x_i(k_i; p_m), \qquad \nu \geq 1. \tag{80}$$

The inequality is strict when the entrusted surplus falls on a positive-measure set; for $\nu > 1$, a strict rise in positive baseline value also makes the concavity channel strict. For $0 < \nu < 1$, no sign is asserted without an additional bound.

Financing-capacity heterogeneity. Within either fixed regime, a finite increase in ℓ_i weakly expands the financeable choice set and therefore weakly raises Ω_i^h and $x_i^{*,h}$. This does not order the reform response. Under the local capital-requirement ordering $J_E < J_I$, the response can be zero below J_E, strictly positive in the financing corridor, and smaller, unchanged, or zero once internalization becomes financeable. Hence no global sign is assigned to the finite-comparison slope of $\Delta x_i(\ell_i)$. At a financing threshold, an ordinary derivative need not exist.

Zero-effect and boundary cases. If the relevant set is null or entrusted advantage is zero, then $\Delta\Omega_i = \Delta x_i = 0$. At $\nu = 1$,

$$x_i^* = \frac{\beta a_i}{\kappa}\Omega_i, \qquad \Delta x_i = \frac{\beta a_i}{\kappa}\Delta\Omega_i,$$

so manufacturing heterogeneity operates entirely through the expected value gain. A high a_i does not create a response when the value gain is zero.

Economic interpretation. Research capability scales how strongly a given increase in the value of advancing projects changes the level of advancement. Weak internal manufacturing capability makes retained outsourcing more likely to add value and, through $J_{I,k} < 0$, makes exposure to the financing corridor more likely at fixed ℓ_i. The resulting response is strongest for high-a_i developers inside the MAH-relevant set; there is no unconditional ordering in ℓ_i. This does not imply more patent applications, basic research, scientific discovery, or breakthrough novelty.

9.4 Corollary: Novelty composition is ambiguous

Assumptions and fixed objects. Fix developer i, financing capacity ℓ_i, and support price p_m. The class $g \in \{O, \text{Inc}\}$ is an exogenous classifier, not a control or state. The same x_i applies to both classes.

Claim and proof. Define $\Omega_{ig}^0 = E_{F_g}[W_i^0(\ell_i)]$ and $\Omega_{ig}^1 = E_{F_g}[\max\{W_i^0(\ell_i), \widetilde{W}_i^E(\ell_i)\}]$. Then

$$\Delta\Omega_{ig} = E_{F_g}\left[\left(\widetilde{W}_i^E - W_i^0\right)_+\right], \qquad \Delta\Omega_i = \sum_{g\in\{O,\text{Inc}\}} \rho_g \Delta\Omega_{ig}. \tag{81}$$

The positive-part identity proves the first equality; the mixture identity for F proves the second. The baseline imposes no restriction ordering the two conditional distributions of entrusted surplus. A support on which only class O has positive entrusted surplus gives $\Delta\Omega_{iO} > \Delta\Omega_{i\text{Inc}}$; interchanging the class supports reverses the ranking. Hence neither ordering follows from the baseline.

Zero-effect and boundary cases. If $\rho_O = 0$, class O contributes zero to the aggregate gain; if $\rho_{\text{Inc}} = 0$, class Inc contributes zero. A conditional class gain may be zero even when the other is positive.

Economic interpretation. Different novelty classes may have different manufacturing requirements, commercial values, success probabilities, costs, or outside options. Only explicit primitive differences can order their responses. No empirical original-versus-incremental ranking is hard-coded into the theory.

9.5 Proposition 4: CMO equilibrium existence and uniqueness

Assumptions and fixed objects. Fix M and all exogenous distributions and primitives. Use the maintained market conditions: continuous weakly decreasing finite demand, continuous strictly increasing supply, positive background demand at zero, vanishing demand at high price, and unbounded high-price supply.

Claim. There exists a unique qualified-capacity clearing price $p_m^*(M) > 0$.

Proof. Define excess demand

$$Z_M(p_m) = D_m(p_m; M) - S_m(p_m). \tag{82}$$

It is continuous. At zero, $Z_M(0) \geq D_m^B(0) > 0$. At sufficiently high prices, total demand tends to zero and supply is unbounded, so $Z_M(p_m) < 0$. The intermediate value theorem gives a positive root. Because demand is weakly decreasing and supply strictly increasing, Z_M is strictly decreasing; the root is unique.

Sufficient conditions and zero-effect cases. Strictly increasing supply is sufficient for uniqueness even when demand has flat regions. At $M = 0$, study-related entrusted demand is zero, but background demand supports a positive pre-MAH price. If background demand at zero were absent, the positive-price conclusion would not follow from these conditions; that is outside the maintained baseline.

Economic interpretation. The capacity price coordinates deterministic entrusted-route demand, project advancement, and supplier capacity through one scalar market. Its existence does not require logit smoothing, supplier entry, or another market.

9.6 Proposition 5: CMO scarcity attenuation

Assumptions and fixed objects. Let $p_m^0 = p_m^*(0)$ and $p_m^1 = p_m^*(1)$. Hold qualified-supply technology, supplier distribution and background demand fixed across regimes. The reform adds nonnegative entrusted-route study demand; old routes remain available.

Proof: equilibrium price comparison. At the pre-MAH price,

$$Z_1(p_m^0) = D_m^{\mathrm{MAH}}(p_m^0; 1) \geq 0,$$

because background demand equals supply at p_m^0. Since Z_1 is strictly decreasing and vanishes at p_m^1,

$$p_m^1 \geq p_m^0, \quad \text{with strict inequality if} \quad D_m^{\mathrm{MAH}}(p_m^0; 1) > 0. \tag{83}$$

Value attenuation. Define the fixed-price and equilibrium-price gains by

$$\Delta\Omega_i^{\mathrm{dir}} = \int [\widetilde{W}_i^E(q, m; 1, p_m^0, \ell_i) - W_i^0(q, m; \ell_i)]_+ dF,$$

$$\Delta\Omega_i^{\mathrm{eq}} = \int [\widetilde{W}_i^E(q, m; 1, p_m^1, \ell_i) - W_i^0(q, m; \ell_i)]_+ dF.$$

Since $W_{i,p_m}^E = -b(m) < 0$ and $p_m^1 \geq p_m^0$, the integrand in the equilibrium expression is pointwise no larger. Both integrands are nonnegative, so

$$0 \leq \Delta\Omega_i^{\mathrm{eq}} \leq \Delta\Omega_i^{\mathrm{dir}}. \tag{84}$$

Advancement attenuation. For a fixed pre-MAH value Ω_i^0, the mapping

$$d \longmapsto \left(\frac{\beta a_i}{\kappa}\right)^{1/\nu} [(\Omega_i^0 + d)^{1/\nu} - (\Omega_i^0)^{1/\nu}]$$

is nonnegative and strictly increasing for $d \geq 0$. Therefore

$$0 \leq \Delta x_i^{\mathrm{eq}} \leq \Delta x_i^{\mathrm{dir}}. \tag{85}$$

Zero-effect and boundary cases. If study demand is zero at p_m^0, the price need not rise. If supply is perfectly elastic at a common $\bar{p}_m$, then $p_m^1 = p_m^0 = \bar{p}_m$ and attenuation disappears. If a high equilibrium price eliminates entrusted advantage for developer i, its equilibrium gain is zero. Even then the old route-set value is preserved: scarcity can attenuate the positive reform gain but cannot make access to the enlarged choice set privately worse.

Economic interpretation. New demand for qualified external capacity bids up its scarcity price. This reduces the value of the entrusted option relative to a fixed-price policy calculation and tempers project advancement. The price response is endogenous; MAH does not directly reduce scarcity or shift supply.

9.7 Proposition 6: Planning-stage and observed outcomes

Assumptions and fixed objects. Fix a regime and its equilibrium objects. The common x_i^* is selected before project characteristics and route assignment. The downstream probabilities $s(q)$ and $s_g(q)$ are exogenous and policy invariant. All integrals use the deterministic route choices evaluated in the same regime.

Planning-stage projects. The equilibrium intensity at which viable projects reach the stage relevant for commercialization is

$$\Lambda_i^{\text{plan}} = a_i x_i^*. \tag{86}$$

This is the planning-stage technology $\lambda_i^{\text{plan}} = a_i x_i$ evaluated at the advancement optimizer.

Observed retained outcomes. Within the paper's original-drug domain, expected realized outcomes for which the developer remains the holder are

$$Y_i^{\text{ret}} = a_i x_i^* \int s(q)\,[\mathbf{1}\{r_i^* = I\} + \mathbf{1}\{r_i^* = E\}]\,dF(q,m). \tag{87}$$

Expected realized outcomes with retained holder–producer separation are

$$Y_i^E = a_i x_i^* \int s(q)\mathbf{1}\{r_i^* = E\}dF(q,m). \tag{88}$$

For $g \in \{O, \text{Inc}\}$, the class contribution to retained outcomes is

$$Y_{ig}^{\text{ret}} = a_i x_i^* \rho_g \int s_g(q)\,[\mathbf{1}\{r_i^* = I\} + \mathbf{1}\{r_i^* = E\}]\,dF_g(q,m). \tag{89}$$

There is one common x_i^*, not an x_{ig} control.

Proof and exact decomposition. Let

$$Q_i^{\text{ret},h} = \int s(q)[\mathbf{1}\{r_i^{*,h} = I\} + \mathbf{1}\{r_i^{*,h} = E\}]dF$$

and let $x_i^h = x_i^*(h, p_m^h)$. Adding and subtracting $a_i x_i^1 Q_i^{\text{ret},0}$ gives

$$\Delta Y_i^{\text{ret}} = a_i(x_i^1 - x_i^0)Q_i^{\text{ret},0} + a_i x_i^1 (Q_i^{\text{ret},1} - Q_i^{\text{ret},0}). \tag{90}$$

The first term is the project-advancement/planning-arrival component. The second is the retained-route composition component. Reassignment from I to E can raise Y_i^E without changing the retained total; movement from T/A to E can raise both.

Zero-effect and ambiguity cases. If advancement and all route indicators are unchanged, observed outcomes are unchanged. A class-specific outcome can have zero change while the other changes. The baseline allows $\Delta Y_{iO}^{\text{ret}} = 0 < \Delta Y_{i\text{Inc}}^{\text{ret}}$ and the reverse; it imposes no ranking. If $\rho_g = 0$, that class contributes zero to the aggregate.

Economic interpretation. The model separates investment in advancing viable projects, their arrival at the commercialization-relevant planning stage, organizational assignment, and realized products. Holder–producer separation is observed only after route assignment. None of these expressions makes patent generation endogenous or lets MAH directly change downstream realization probabilities.

9.8 Scope conclusion

The six propositions and the novelty corollary exhaust the baseline results established in this section. They add no market, no entry margin, no welfare object, no logit route share, no class-specific advancement control, and no direct policy effect on research capability, project draws, manufacturing technology, capacity supply, or downstream realization.

10 Comparative Statics and Boundary Cases

This section distinguishes four operations that cannot be interchanged:

1. a derivative with respect to a continuous primitive at a fixed candidate CMO price p_m;
2. a derivative with respect to a continuous primitive after the CMO market clears at p_m^*;
3. a finite comparison between the binary regimes $M = 0$ and $M = 1$;
4. a limiting-case calculation.

There is no derivative with respect to binary M. Before each economic interpretation, the mathematical margin is named as commercial return, commercialization-route choice, project advancement, the CMO price, or realized product output. Upstream research and patent generation are not endogenous margins of the baseline.

10.1 Commercial pricing and return

Object and held-fixed variables. The differentiated objects are the optimized product price, operating profit, and commercial present value. Project value q, marginal manufacturing cost c, and the demand and discount primitives are held fixed except for the single argument named in each derivative. These are conditional product-market calculations; neither M nor p_m^* is varied.

The pricing solution and operating profit derived above can be written as

$$p^*(c) = \frac{\varepsilon}{\varepsilon - 1}c, \qquad \pi(q, c) = \frac{Aq}{\varepsilon}\left(\frac{\varepsilon}{\varepsilon - 1}c\right)^{1-\varepsilon}.$$

Direct differentiation gives

$$\frac{\partial p^*}{\partial c} = \frac{\varepsilon}{\varepsilon - 1} > 1, \qquad \frac{\partial \pi}{\partial q} = \frac{\pi}{q} > 0, \qquad \frac{\partial \pi}{\partial c} = (1 - \varepsilon)\frac{\pi}{c} < 0. \tag{91}$$

Since $R(q, c) = \pi(q, c)/(1 - \beta\varphi)$,

$$R_q = \frac{R}{q} > 0, \qquad R_c = (1 - \varepsilon)\frac{R}{c} < 0, \tag{92}$$

and, holding π fixed,

$$R_\beta = \frac{\varphi\pi}{(1 - \beta\varphi)^2} \geq 0, \qquad R_\varphi = \frac{\beta\pi}{(1 - \beta\varphi)^2} > 0.$$

The first inequality is an equality when $\varphi = 0$.

Sufficient conditions, dimensions, and zero effects. The signs use $A, q, c > 0$, $\varepsilon > 1$, $\beta \in (0, 1)$, and $\varphi \in [0, 1)$. The price derivative is dimensionless; R_q and R_c have currency per unit of their respective arguments. A change in β has no continuation-value effect when $\varphi = 0$. None of these derivatives is a policy effect unless a specified route technology changes the cost argument supplied to R.

Economic interpretation and empirical prediction. This is a commercial-return margin. Higher manufacturing cost is passed through to the product price and lowers optimized commercial value, while a higher project-value shifter raises it. The empirical prediction is conditional cost pass-through and lower commercial return for more costly manufacturing, not a claim about patenting or upstream research.

10.2 Commercialization-route choice at fixed CMO price

Object and held-fixed variables. The object is the internal-minus-entrusted value gap and its conditional capability cutoff. Hold (q, m), all technologies and outside options fixed. For capability derivatives also hold finite (τ_E, p_m) fixed. For each implicit cutoff derivative, vary only the continuous argument displayed.

Conditional on both retained routes being financeable,

$$\Delta_{IE,k} = s(q)R_c(q, c_I)c_{I,k} - F_{I,k} > 0. \tag{93}$$

The operating-cost component is weakly positive and may be zero when $s(q) = 0$; the setup-cost component is strictly positive. Under the cell-specific crossing condition, the implicit-function theorem gives

$$\left.\frac{\partial k^*}{\partial \tau_E}\right|_{q,m,p_m} = -\frac{1}{\Delta_{IE,k}(k^*)} < 0, \qquad \left.\frac{\partial k^*}{\partial p_m}\right|_{q,m,\tau_E} = -\frac{b(m)}{\Delta_{IE,k}(k^*)} < 0. \tag{94}$$

Both are continuous-argument derivatives inside the finite-wedge domain. They are not derivatives with respect to M, and the second is not an equilibrium-price effect.

Separately, $J_{I,k} < 0$ means that stronger internal capability weakly expands the set $\{(m, \ell_i) : J_I(m, k_i) \leq \ell_i\}$. This is a financing-feasibility comparison, not part of the value-gap derivative.

Binary policy comparison. Let $W_i^0(\ell_i) = \max\{\widetilde{W}_i^I(\ell_i), W^T, 0\}$. At any fixed finite p_m,

$$\widetilde{W}_i(q, m; 1, p_m, \ell_i) - \widetilde{W}_i(q, m; 0, p_m, \ell_i) = [\widetilde{W}_i^E(q, m; 1, p_m, \ell_i) - W_i^0(q, m; \ell_i)]_+ \geq 0. \tag{95}$$

This finite comparison adds an option without changing the old route values. It is strict only on $\mathcal{C}_i(p_m)$.

Sufficient conditions and zero effects. The cutoff signs require internal feasibility, the maintained technology signs, finite (τ_E, p_m), strict $\Delta_{IE,k} > 0$, and endpoint crossing. If crossing fails, no interior cutoff is asserted. If T or A dominates, the I/E cutoff does not determine the realized route. If $J_E > \ell_i$ or $W_i^E \leq W_i^0$, the finite policy gain is zero.

Economic interpretation and empirical prediction. This is a commercialization-route margin. Better internal manufacturing capability makes internal production relatively more attractive; a lower finite entrusted-route burden expands the capability region using E, while a higher fixed CMO price contracts it. The empirical prediction is sorting of holder–manufacturer organization by predetermined manufacturing capability and project requirement, conditional on outside options.

10.3 Project advancement at fixed CMO price

Object and held-fixed variables. The object is the unique common project-advancement intensity

$$x_i^*(M, p_m) = \left[\frac{\beta a_i \Omega_i(M, p_m)}{\kappa}\right]^{1/\nu}.$$

Each partial derivative below holds M, p_m, ν and all unlisted primitives fixed. In particular, route-value objects are held fixed when a_i or κ is varied. The formulas apply at $\Omega_i > 0$; the $\Omega_i = 0$ corner is handled separately.

Differentiation of the advancement FOC gives

$$\frac{\partial x_i^*}{\partial \Omega_i} = \frac{x_i^*}{\nu \Omega_i} > 0, \qquad \frac{\partial x_i^*}{\partial a_i} = \frac{x_i^*}{\nu a_i} > 0, \qquad \frac{\partial x_i^*}{\partial \kappa} = -\frac{x_i^*}{\nu \kappa} < 0. \tag{96}$$

The corresponding elasticities are $1/\nu, 1/\nu, -1/\nu$. Because the optimized project value obeys the deterministic envelope, $\Omega_{i,p_m} = -\chi_i^E$ almost everywhere, so

$$\left.\frac{\partial x_i^*}{\partial p_m}\right|_M = -\frac{x_i^*}{\nu\Omega_i}\chi_i^E \le 0. \tag{97}$$

This is a fixed-candidate-price derivative and already includes both the intensive value effect and deterministic route-envelope effect inside Ω_i.

For binary M, define at a common support price

$$\Delta\Omega_i(p_m) = \int [\widetilde{W}_i^E(q, m; 1, p_m, \ell_i) - W_i^0(q, m; \ell_i)]_+ \, dF.$$

The correct policy calculation is the finite difference

$$\Delta x_i(p_m) = \left(\frac{\beta a_i}{\kappa}\right)^{1/\nu} \left[(\Omega_i^0 + \Delta\Omega_i(p_m))^{1/\nu} - (\Omega_i^0)^{1/\nu}\right] \ge 0. \tag{98}$$

It is strict if and only if $\Delta\Omega_i(p_m) > 0$.

Sufficient conditions and zero effects. The partials require $\Omega_i > 0$. At the zero-value corner, $x_i^* = 0$; global monotonicity follows directly from the closed form. Higher a_i scales a positive response but cannot create one when the entrusted surplus is zero. The weak price effect is zero when no project uses E, so $\chi_i^E = 0$. The manufacturing-capability ordering of the level response additionally uses $\nu \ge 1$; no such sign is claimed for $0 < \nu < 1$ without another bound.

Economic interpretation and empirical prediction. This is development-stage project advancement, not upstream research. Expected commercialization value raises the return to advancing viable projects; higher advancement productivity amplifies a positive value gain, whereas a steeper cost scale reduces the chosen intensity. The prediction is a larger project-advancement response among high-a_i developers for which entrusted manufacturing adds value, especially when internal manufacturing capability is weak under the stated curvature condition. No patent-application response follows mechanically.

10.4 CMO supply, demand, and continuous equilibrium derivatives

Supplier object and held-fixed variables. For a supplier with $p_m > 0$, vary only candidate price or efficiency while holding the other argument and the cost function fixed. Implicit differentiation of $p_m = \Psi_s(s_j^*; z_j)$ gives

$$\frac{\partial s_j^*}{\partial p_m} = \frac{1}{\Psi_{ss}} > 0, \qquad \frac{\partial s_j^*}{\partial z_j} = -\frac{\Psi_{sz}}{\Psi_{ss}} > 0. \tag{99}$$

These are capacity-supply derivatives; policy does not shift Ψ or z_j.

Study-demand object. At differentiability points of the aggregated deterministic-choice functions,

$$\frac{\partial D_m^{\mathrm{MAH}}}{\partial p_m} = \int a_i \left[\frac{\partial x_i^*}{\partial p_m}\chi_i^E + x_i^* \frac{\partial \chi_i^E}{\partial p_m}\right] dH(a, k, \ell) \le 0. \tag{100}$$

The first term is the project-value/advancement response; the second is the commercialization-route response. Where a classical derivative does not exist, the global weak decrease follows from monotonicity and atomless aggregation.

Continuous equilibrium perturbation. Consider an infinitesimal continuous, non-policy perturbation. Let dD_m and dS_m denote its direct demand and supply shifts at a fixed price. Hold every other primitive fixed. Differentiating $Z(p_m) = D_m(p_m) - S_m(p_m) = 0$ at the unique equilibrium gives

$S_{m,p} - D_{m,p} > 0$ and

$$dp_m^* = \frac{dD_m - dS_m}{S_{m,p} - D_{m,p}}. \tag{101}$$

A demand-increasing shifter raises the equilibrium price when supply is not shifted; a supply-increasing shifter lowers it when demand is not shifted. This formula is not applied to binary M.

Sufficient conditions, dimensions, and zero effects. The supplier signs use $\Psi_{ss} > 0$ and $\Psi_{sz} < 0$. The aggregate price derivative uses differentiability and the same strict excess-demand slope that ensures uniqueness. It is zero when the perturbation changes neither demand nor supply at the equilibrium. Equation (101) is dimensionally a change in CMO price.

Interpretation and prediction. These are CMO capacity and equilibrium-price margins. A higher capacity price induces more qualified supply but reduces study demand through both fewer entrusted assignments and weaker project advancement. The prediction is an upward-sloping supply response and a scarcity price that responds positively to continuous demand pressure, conditional on unchanged supply technology.

10.5 Binary reform at the equilibrium CMO price

Object and finite comparison. Let $p_m^h = p_m^*(h)$ for $h \in \{0, 1\}$. Supply technology, the supplier distribution, and background demand are held fixed. Since $D_m^{\mathrm{MAH}}(p_m; 0) = 0$, market clearing at p_m^0 implies

$$Z_1(p_m^0) = D_m^{\mathrm{MAH}}(p_m^0; 1) \geq 0.$$

Strict monotonicity of Z_1 therefore gives the finite comparison

$$p_m^1 \geq p_m^0, \qquad p_m^1 > p_m^0 \iff D_m^{\mathrm{MAH}}(p_m^0; 1) > 0. \tag{102}$$

No derivative with respect to M is used.

Define the fixed-price direct gain and the equilibrium gain as

$$\Delta\Omega_i^{\mathrm{dir}} = \int [\widetilde{W}_i^E(q, m; 1, p_m^0, \ell_i) - W_i^0(q, m; \ell_i)]_+ \, dF,$$

$$\Delta\Omega_i^{\mathrm{eq}} = \int [\widetilde{W}_i^E(q, m; 1, p_m^1, \ell_i) - W_i^0(q, m; \ell_i)]_+ \, dF.$$

Because $W_{i,p_m}^E = -b(m) < 0$, the price order yields

$$0 \leq \Delta\Omega_i^{\mathrm{eq}} \leq \Delta\Omega_i^{\mathrm{dir}}. \tag{103}$$

Applying the strictly increasing advancement mapping to these value gains gives

$$0 \leq \Delta x_i^{\mathrm{eq}} \leq \Delta x_i^{\mathrm{dir}}. \tag{104}$$

Conditions and zero effects. The price order requires policy-invariant supply/background demand and nonnegative post-reform study demand. Strictness requires positive study demand at p_m^0. A developer's equilibrium gain is zero when the entrusted option remains below its old-route maximum at p_m^1. Scarcity can eliminate an incremental gain but cannot lower the old choice-set value.

Economic interpretation and empirical prediction. This is an equilibrium CMO-price feedback on project advancement. New entrusted-route demand bids up scarce qualified capacity, so a fixed-price calculation overstates the equilibrium value and advancement response. The prediction is a weak post-reform increase in the CMO capacity price when the study cohort creates positive demand and a smaller project response in markets with less elastic capacity supply.

10.6 Planning-stage and realized-product finite comparisons

Objects and held-fixed primitives. The objects are planning-stage project intensity and realized retained-product outcomes, evaluated at the regime-specific equilibrium. Predetermined a_i, project distributions, realization probabilities, technologies and outside options are held fixed. The regime changes only route availability and the endogenous CMO price.

Planning-stage intensity satisfies the exact finite difference

$$\Delta\Lambda_i^{\text{plan}} = a_i \Delta x_i^{\text{eq}} \geq 0. \tag{105}$$

Let

$$Q_i^{\text{ret},h} = \int s(q)[\mathbf{1}\{r_i^{*,h} = I\} + \mathbf{1}\{r_i^{*,h} = E\}]\, dF.$$

Adding route E cannot remove an old option. If E wins, it is retained; if it does not win, the old maximizer is unchanged. Hence $Q_i^{\text{ret},1} \geq Q_i^{\text{ret},0}$. The exact decomposition is

$$\Delta Y_i^{\text{ret}} = a_i(x_i^1 - x_i^0)Q_i^{\text{ret},0} + a_i x_i^1(Q_i^{\text{ret},1} - Q_i^{\text{ret},0}) \geq 0. \tag{106}$$

The first term is project advancement; the second is retained-route composition. The same decomposition applies within class g using $\rho_g F_g$ and the common x_i, not a class-specific control.

Sufficient conditions and zero effects. A strict planning-stage increase requires a positive equilibrium expected-value gain. A strict retained-product increase requires a positive advancement term or positive-measure movement from T/A into E with positive realization weight. Reassignment from I to E raises observed holder–producer separation but, by itself, does not change the retained total. If advancement and route indicators are unchanged, every displayed outcome change is zero.

Economic interpretation and empirical prediction. These are planning-stage and realized-product margins. The model predicts that retained holder–producer separation can rise after route E becomes available, and separates that assignment change from the number of viable projects advanced and from downstream realization. Product-level holder and manufacturer records can test the assignment prediction. Patent applications are not an outcome in these expressions.

10.7 Mandatory limiting cases

Internal capability $k_i \to \infty$. On the feasible domain, $\Delta_{IE,k} > 0$, so increasing capability weakly favors I relative to E. Under the maintained upper-end crossing condition, $\Delta_{IE} > 0$ for sufficiently large k_i. This is a relative route-value statement; T or A may still dominate both.

Low internal capability and financing. Low k_i raises internal operating cost, setup cost, and the up-front commitment $J_I(m, k_i)$, but it does not make internalization technologically impossible. If $J_I(m, k_i) > \ell_i$, route I is financially infeasible; if ℓ_i is sufficiently high, the same low-k_i developer can still internalize. The old hard-infeasibility approximation is recovered only as the limit $J_I(m, k_i) \to +\infty$.

Financing-capacity regions. Under the local ordering $J_E(m) < J_I(m, k_i)$, $\ell_i < J_E$ makes both retained routes infeasible, $J_E \leq \ell_i < J_I$ is the financing corridor in which only E is financeable, and $\ell_i \geq J_I$ restores I as an old option. Hence the binary MAH effect can be zero–positive–smaller as ℓ_i crosses the two thresholds; it is not globally monotone in financing capacity. Within a fixed regime, however, optimized value is weakly increasing in ℓ_i because the feasible route set only expands.

Vanishing and abundant financing. As $\ell_i \to 0$, both retained routes may be infeasible and MAH can have zero effect. If $\ell_i \geq \max\{J_I(m, k_i), J_E(m)\}$ for all relevant pairs, financing never binds and $\widetilde{W}_i^I = W_i^I$, $\widetilde{W}_i^E = W_i^E$; the previous no-financing-friction model is nested.

Entrusted financing requirement. The limit $J_E \to 0$ is an extreme low-capital outsourcing boundary, not the baseline. If $J_E \geq J_I$, route E has no financing-relief advantage over internalization for that pair, although it may still add value through its organizational technology and cost structure.

Pre-MAH regime. When $M = 0$, $\tau_E(0) = +\infty$, so $W_i^E = -\infty$, $\chi_i^E = 0$, and $D_m^{\mathrm{MAH}} = 0$. The old route set and background CMO market remain.

Infinite CMO price. Because $b(m) > 0$,

$$\lim_{p_m \to \infty} W_i^E(q, m; 1, p_m) = -\infty.$$

Thus E ceases to be optimal and

$$\lim_{p_m \to \infty} \widetilde{W}_i(q, m; 1, p_m, \ell_i) = W_i^0(q, m; \ell_i). \tag{107}$$

The newly added option disappears economically but cannot reduce the old route-set value.

Zero entrusted-route advantage. If $J_E(m) > \ell_i$ or $W_i^E(q, m; 1, p_m) \leq W_i^0(q, m; \ell_i)$ for every draw, then

$$\Delta\Omega_i(p_m) = 0, \qquad \Delta x_i(p_m) = 0. \tag{108}$$

At the equilibrium price, the same conclusion applies when the inequality holds at p_m^1.

Perfectly elastic CMO supply. If qualified capacity is available at the common administered/competitive price $\bar{p}_m$ in both regimes, then

$$p_m^1 = p_m^0 = \bar{p}_m, \qquad \Delta\Omega_i^{\mathrm{eq}} = \Delta\Omega_i^{\mathrm{dir}}, \qquad \Delta x_i^{\mathrm{eq}} = \Delta x_i^{\mathrm{dir}}.$$

The equilibrium attenuation channel disappears. This is a boundary outside the strictly increasing-supply condition used for uniqueness, not a claim that baseline supply is perfectly elastic.

Quadratic advancement cost. For $\nu = 1$,

$$x_i^*(M, p_m) = \frac{\beta a_i}{\kappa} \Omega_i(M, p_m), \qquad \Delta x_i = \frac{\beta a_i}{\kappa} \Delta\Omega_i. \tag{109}$$

The value response is exactly linear.

Novelty-share boundaries and no fixed ranking. The common-control decomposition is

$$\Delta\Omega_i = \rho_O \Delta\Omega_{iO} + \rho_{\mathrm{Inc}} \Delta\Omega_{i\mathrm{Inc}}. \tag{110}$$

If $\rho_O = 0$, the original-oriented class contributes zero to the aggregate; if $\rho_{\mathrm{Inc}} = 0$, the incremental class contributes zero. Either conditional gain may be zero while the other is positive. Reversing the class supports of positive entrusted surplus reverses their ordering, so no fixed ranking follows from the baseline.

10.8 Comparative-static scope conclusion

All central signs above follow from the maintained primitive, technology, market, and mapping assumptions and the deterministic option-value structure. This section adds no route, control, state, market, welfare object, patent outcome, logit share, inclusive value or continuous implementation index. Its finite policy chain is

$$\begin{aligned} M &\longrightarrow \text{availability of retained entrusted manufacturing} \longrightarrow \Omega_i \\ &\longrightarrow x_i^* \longrightarrow \text{planning, route, and realized-product objects.} \end{aligned}$$

with the endogenous CMO price attenuating, rather than directly causing, the project-value response.

11 Observed Outcomes and the Data Boundary

The required propositions derived planning-stage and observed-product outcomes. This section fixes their interpretation and timing. It adds no primitive, control, realization channel, market, or empirical identification claim.

11.1 Outcome hierarchy

The baseline preserves the strict conceptual distinction

$$\boxed{\text{upstream research} \neq \text{project advancement} \neq \text{commercialization organization} \neq \text{realized product.}} \tag{111}$$

Here x_i is original-drug innovation investment or project-advancement intensity. It is broader than pure clinical-development effort but is not patent applications, basic research, or upstream scientific discovery. The optimal planning-stage arrival intensity is $\Lambda_i^{\text{plan}} = a_i x_i^*$; route choice then assigns a realized project to I, E, T, or A; only afterward can a product be realized and holder–producer separation be observed.

11.2 Anticipated and realized sequences

The ex ante policy channel is

$$M \longrightarrow \text{anticipated availability and finance-adjusted value of } E \longrightarrow \Omega_i \longrightarrow x_i^*. \tag{112}$$

The later realized sequence is

$$x_i^* \longrightarrow \Lambda_i^{\text{plan}} \longrightarrow r_i^* \longrightarrow \text{observed holder–producer separation} \longrightarrow \text{realized products.} \tag{113}$$

Observed separation is therefore a downstream route outcome, not an event that precedes or causes the ex ante advancement choice. Predetermined ℓ_i determines which retained organizational commitments can be funded, but MAH does not change ℓ_i, J_I, or J_E.

11.3 Novelty classes under one common control

For $g \in \{O, \text{Inc}\}$, the class-specific retained outcome retains the form established in Proposition 6:

$$Y_{ig}^{\text{ret}} = a_i x_i^* \rho_g \int s_g(q) \left[\mathbf{1}\{r_i^* = I\} + \mathbf{1}\{r_i^* = E\}\right] dF_g(q, m). \tag{114}$$

The classifier changes the conditional distribution used in the integral; it does not create a control x_{ig}. The baseline permits either class to have a zero response while the other responds, and imposes no ranking between their reform gains.

11.4 Measurement boundary

Patent applications measure a patenting margin that may reflect upstream research activity, but they are not direct observations of upstream research and are not baseline endogenous outcomes. An isolated patent decline does not by itself falsify the mechanism, and the revised model assigns no patent sign. Activity around an IND or another regulatory application, and early clinical milestones, may proxy project advancement only when their measurement definition and timing are validated. Product-level holder–manufacturer separation may proxy route E only when holder identity, manufacturer identity, product, and date are aligned. Approval or launch data are realized-product outcomes, not direct observations of x_i.

Approval-side and product-side data generally do not separately identify

$$\{x_i,\ a_i,\ k_i,\ \ell_i,\ J_I,\ J_E,\ s(q),\ \tau_E,\ \mu_E,\ p_m\}\,. \tag{115}$$

They may discipline composite objects such as planning-stage intensity, retained-route realization, or holder–producer separation. A primitive-complete theory does not imply primitive-by-primitive empirical identification.

11.5 Scope conclusion

The formal distinction is therefore among original-drug innovation investment/project advancement, route-planning-stage arrival, commercialization organization, and observed approval, launch, or retained product. Patent generation is not added as a baseline outcome, and downstream realization probabilities remain policy invariant. No empirical data set or proxy is asserted to be available here.

12 Financing, Research, and Patent Boundaries of the Baseline

MAH directly changes downstream commercialization options through the institutional wedge $\tau_E(M)$, whose infinite/finite value encodes the effective availability of retained entrusted manufacturing. Predetermined financing capacity ℓ_i determines whether a developer can fund the up-front organizational commitment J_I or J_E, but MAH changes none of these three objects. The common control x_i remains original-drug innovation investment / project-advancement intensity. It is broader than pure clinical effort, but excludes upstream patent-generating research, basic-compound discovery, and patent applications. Accordingly, the baseline permits

$$M \longrightarrow \text{anticipated availability and finance-adjusted value of } E \longrightarrow \Omega_i \longrightarrow x_i^*,$$

but it does not imply that MAH raises patent applications.

This distinction is empirically relevant. Gu (2024) documents that the reform permitting production outsourcing raises clinical-trial registrations among innovators without production facilities (Table II, p. 45), while patent applications decline among firms with fewer financial resources (Tables III–IV, pp. 46–47) and do not significantly decline for firms with more resources. Granted- and core-patent proxies do not show a statistically significant decline (Table V, p. 48). These findings show that clinical-trial and patenting margins can diverge; patent outcomes are not direct observations of upstream research. Gu studies financing-induced reallocation between research and development. The present baseline instead lets financing affect whether vertical integration or retained outsourcing can be funded. MAH changes the capital intensity of the organizational route needed to retain rights; it does not relax financing constraints or allocate resources between research and development.

The same paper reports that the trial response is concentrated in the first stage (Table VIII, p. 51) and in incrementally innovative chemical drugs rather than originally innovative chemical drugs, while originally innovative biological drugs show a smaller positive point estimate (Table VII, p. 50). The author describes these cross-class patterns as more pronounced for incremental projects, but no formal cross-column equality test is reported. They therefore motivate heterogeneity tests; they are not restrictions imposed on the present baseline. The common x_i, exogenous shares ρ_g, and conditional distributions F_g remain compatible with either class responding more, less, or not at all.

Gu (2024) also finds that pure researchers become more likely to initiate a first clinical trial and enter development (Table IX, p. 52). That result is an extensive-margin transition outside the present fixed-population partial equilibrium and may motivate a separately specified entry extension. The present model does not claim novelty over Gu until an empirical implementation shows that predetermined financing capacity shifts retained holder–manufacturer separation through the organizational-feasibility channel.

13 Explicit Extensions: Not the Baseline

Separation rule. Every object in this section is inactive in the baseline. None enters baseline symbols, assumptions, equilibrium, or propositions. All blocks below remain documented possibilities. A former research–development allocation extension has been retired from the active paper because the financing revision deliberately preserves one common advancement control and does not import a Gu-style resource-allocation mechanism.

13.1 Extension 1: Smooth route choice

If a future quantitative application requires smooth route shares, it may replace the local deterministic utility comparison by

$$U_{ir} = W_{ir} + \epsilon_{ir}, \qquad r \in \{I, E, T, A\}.$$

Under an independently justified iid Type-I extreme-value distribution with scale $\sigma > 0$, the extension-only probability would be

$$P_{ir}^{\text{logit}} = \frac{\exp(W_{ir}/\sigma)}{\sum_{\ell \in \{I,E,T,A\}} \exp(W_{i\ell}/\sigma)}. \tag{116}$$

This formula is not used by any baseline demand, cutoff, proposition, or equilibrium equation. Activation requires route-share data, a quantitative need, and shock-scale identification. Deterministic r_i^* remains the theoretical baseline.

13.2 Extension 2: Route-specific implementation

If later evidence supports route-specific post-assignment implementation probabilities, an extension may introduce $\chi^I(q,m)$ and $\chi^E(q,m)$. These would replace neither the baseline route choice nor the policy-invariant downstream probability without a new institutional model. In particular,

$$M = 1 \;\not\Rightarrow\; \chi^E(q,m) \uparrow$$

by assumption. Activation requires an operational definition, route-aligned data, and a causal justification for any policy dependence.

13.3 Extension 3: Transfer-market microfoundation

A future adoption-cost, search, or bargaining block may microfound the outside value $T(q,m)$. No bargaining protocol, transfer price, adoption cost, or counterparty choice is selected here. Until a separately specified design specifies agents, information, surplus, outside options and timing, $T(q,m)$ remains the exogenous non-retained outside option of the baseline.

13.4 Extension 4: Dynamic firm evolution

A recursive model is admissible only when a future question and evidence support a genuine state transition, such as accumulated manufacturing capability, a persistent product portfolio, entry/exit, or transition from a research-oriented developer to an integrated incumbent. No Bellman equation is written here because no state vector or transition law has been specified. This prevents static continuation-value accounting from being relabeled as a dynamic foundation.

13.5 Extension 5: Multi-CMO matching

A richer search or matching model may be considered if match-level data identify developers, suppliers, technology compatibility, search duration, prices and match timing. No matching function, bargaining rule or congestion externality is imposed here. The baseline continues to clear one aggregate qualified-capacity market at p_m^*.

13.6 Extension 6: Research-versus-development allocation

This block is inactive. A future specification could use upstream research, project development, a common resource ceiling, and an explicit patent-production equation, but none is part of the revised paper. In particular, the baseline has no constraint of the form $x_i^R + x_i^D \leq B_i$, no research-to-development reallocation, and no predicted patent response.

The baseline continues to retain one common x_i, commercialization organization, retained authorization, manufacturing-capability matching, and CMO scarcity. No baseline result cites x_i^R, x_i^D, the resource ceiling, or the patent-production equation.

13.7 Isolation conclusion

All optional blocks remain inactive. Conversely, the baseline does not depend on any extension-only symbol or equation. Predetermined financing capacity ℓ_i is a baseline characteristic, but borrowing, debt contracts, VC bargaining, bank screening, endogenous interest rates, and finance-market clearing remain outside the model.

References

Gu, Shi, "Production Outsourcing and Innovation: Evidence from China's Pharmaceutical Industry," Working Paper SSRN 4770849, Kellogg School of Management, Northwestern University 2024. August 22, 2024 version.